Miguel Monteiro

Mechanical Engineer

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Education

MSc in Aeronautical and Vehicular Structures, Faculty of Engineering of the University of Porto *Present*
BSc in Mechanical Engineering, Faculty of Engineering of the University of Porto *Present*

Work Experience

Formula Student FEUP

Porto, Portugal

Head of Chassis

Jan. 2025 – Present

- Managed **3 subsystems** and **8 people**;
- Planned the team's **first monocoque** as a proof-of-concept;
- Designed the third spaceframe achieving a mass of 29 kg, a **24% decrease**;
- Presented all static disciplines in 2025 competitions, achieving **6 podiums** out of 9, including the **first Portuguese podium at Formula Student Germany** in the **EV** category;
- **Co-coordinated** 2 Master and 1 Bachelor thesis about **composite design** and **FEM simulations**.

Primary Structure Manufacturing Responsible

Sep. 2024 – Jan. 2025

- Coordinated the chassis' production and obtained **6 new partners** to ensure project feasibility;
- Designed the needed infrastructure to guarantee welding tolerance.

Primary Structure Engineer

Nov. 2023 – Sep. 2024

- Designed the spaceframe and achieved a mass of 38 kg, a **25% reduction**;
- Created a **topology optimized** steering support using SLM, achieving a **55% reduction** from its machined solution;
- Modeled and sized all primary mounts using both **analytical and computational methods**;
- **Co-coordinated** a Bachelor thesis about the **mechanical characterization of sandwich panels**.

Adamastor Supercars

Porto, Portugal

Research Internship

Jan. 2025 – Jun. 2025

- Bachelor thesis about the influence of inserts on the stiffness of sandwich panels, achieved a **grade of 18/20**;
- **Developed** a **numerical sizing tool** for through-the-core inserts;
- Performed **linear simulations** to predict the **elastic response** of **sandwich panels**;
- Planned, manufactured and tested specimens under **four-point bending**.

Additional Experience

NEEM FEUP

Porto, Portugal

Marketing Member

Feb. 2023 – Feb. 2024

- Responsible for clothing design and manufacture;
- Photography and creation of media content for the National Meeting of Mechanical Engineers.

Skills

Languages	**Portuguese** (native), **English** (C2, proficient), **German** (B1, intermediate)
CAD	Siemens NX, Dassault Solidworks, Dassault Catia, Autodesk Inventor, Autodesk Fusion
FEM	Abaqus, Ntopology
Programming	Matlab, Python, Javascript
Management	Notion, Gantt